"Negative Viscosity" in a Two-Dimensional Magnetic Fluid: the Complete Theory Re-derived from First Principles in 2D

The exact analog, for planar rotors in a planar flow, of the theory chain Shliomis (1972) → Martsenyuk–Raikher–Shliomis (1974) → Shliomis–Morozov (1994) → Bacri et al. (1995)

Derivation notes. Every boxed result is derived from the 2D microscopics and checked (limits, identities, or numerics against the *exactly solvable* 2D Fokker–Planck equation); see the Verification Ledger, Sec. 13.

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1 The 2D model: definition, realizations, and honest caveats

Model. Everything lives in the xy plane:

- Rigid magnetic dipoles whose moments $\mathbf{m} = m\,\mathbf{e}(\phi)$, $\mathbf{e} = (\cos\phi, \sin\phi)$, are confined to the plane; the *single* rotational degree of freedom is the angle ϕ (rotation axis $\hat{\mathbf{z}}$). Orientation space is the circle S^1 (in 3D it was the sphere S^2).
- Incompressible 2D flow $\mathbf{v}(x, y)$; vorticity is the scalar $\Omega = \frac{1}{2}(\partial_x v_y - \partial_y v_x)$, i.e. $\mathbf{\Omega} = \Omega\hat{\mathbf{z}}$.
- In-plane field $\mathbf{H} = H(t)\mathbf{h}$, $\mathbf{h} = \hat{\mathbf{x}}$; for the Bacri problem $H(t) = H_0 \cos\omega t$.
- η = 2D shear viscosity, φ = *area* fraction of disks of radius a, n = number per area, $\varphi = n\pi a^2$.

Physical realizations (and what changes in each):

1. *Strict 2D / long parallel cylinders.* Magnetically hard rods with transverse moments; all quantities per unit length. Rotational drag is the 2D one derived in Sec. 2 (rotation, unlike translation, has *no* Stokes paradox in 2D, so the theory is internally consistent).
2. *Membrane-embedded disks (Saffman–Delbrück setting).* The rotational drag $\zeta_r = 4\pi\eta_m a^2$ (η_m = membrane surface viscosity) is again finite and local; the results below apply with $\eta \to \eta_m$.
3. *Quasi-2D: a monolayer of 3D spheres with easy-plane moments.* The drag is the 3D Stokes one, $\zeta_r = 8\pi\eta_{3\mathrm{D}} a^3$ per particle; only the *orientational statistics* are 2D. All formulas below hold with that ζ_r; note the important consequence $\tau_B^{2\mathrm{D}} = \zeta_r/kT = 2\tau_B^{3\mathrm{D}}$ (Sec. 2), so the sign-change frequency is *half* the 3D one for the same particle.

Caveat (do not confuse with Hele–Shaw). Depth-averaged flow between close plates is *not* 2D Navier–Stokes (it is Darcy-like, with wall friction $\propto v$ dominating $\eta\Delta\mathbf{v}$); the viscosity-renormalization theorem of Sec. 5 does not transfer to that geometry without modification. Also, at fixed φ dipole–dipole interactions are relatively stronger in 2D (chaining in-plane), so the dilute single-particle theory has a narrower window of validity than in 3D.

Part I
Statistical mechanics of a planar rotor

2 Single-particle physics in 2D

2.1 Rotational friction of a disk: $\zeta_r = 4\pi\eta a^2$ (derived)

A disk of radius a rotating at ω in a quiescent 2D Stokes fluid generates the purely azimuthal flow

$$v_\theta(r) = \frac{\omega a^2}{r}, \tag{1}$$

which satisfies 2D Stokes exactly: the azimuthal Stokes operator is $\frac{\mathrm{d}}{\mathrm{d}r}\left[\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}(rv_\theta)\right]$, and $rv_\theta = \omega a^2 =$ const makes it vanish identically, with $p =$ const. (Contrast translation, where no decaying 2D Stokes solution exists — the Stokes paradox; rotation is safe.) The stress

$$\sigma_{r\theta} = \eta\, r\frac{\mathrm{d}}{\mathrm{d}r}\Big(\frac{v_\theta}{r}\Big) = \eta r\Big(-\frac{2\omega a^2}{r^3}\Big) = -\frac{2\eta\omega a^2}{r^2}, \tag{2}$$

gives the torque exerted by the fluid on the disk

$$T = \oint_{r=a} a\,\sigma_{r\theta}\,\mathrm{d}\ell = 2\pi a\cdot a\cdot\Big(-\frac{2\eta\omega a^2}{a^2}\Big) = -4\pi\eta a^2\omega \;\Rightarrow\; \boxed{\zeta_r = 4\pi\eta a^2}, \tag{3}$$

and hence the friction *density*

$$\boxed{n\zeta_r = 4\eta\,(n\pi a^2) = 4\eta\varphi} \qquad (\text{2D analog of } n\zeta_r = 6\eta\varphi \text{ in 3D}). \tag{4}$$

Inertial time: $I_1 = \frac{1}{2}\rho_s\pi a^4$ (disk), so $\tau_s = I_1/\zeta_r = \rho_s a^2/8\eta$ — as negligible as in 3D ($\tau_s/\tau_B \sim 10^{-5}$ for colloidal scales). All inertia-elimination steps below are therefore identical to 3D.

2.2 Brownian time: $\tau_B = 1/D_r$ (*not* $1/2D_r$)

Angular diffusion on the circle: $\langle \delta\phi^2 \rangle = 2D_r t$, $D_r = kT/\zeta_r$. The orientational harmonics $e^{in\phi}$ relax at the rate $n^2 D_r$ (proved as part of Sec. 3); the magnetization is the $n = 1$ harmonic, so

$$\boxed{\tau_B^{\mathrm{2D}} = \frac{1}{D_r} = \frac{\zeta_r}{kT}} \qquad \text{vs.} \qquad \tau_B^{\mathrm{3D}} = \frac{1}{2D_r} \tag{5}$$

(in 3D the $l = 1$ spherical harmonic relaxes at $l(l+1)D_r = 2D_r$). Same particle, same drag: the 2D relaxation is *twice slower*. For the strictly-2D disk, $\tau_B = 4\pi\eta a^2/kT$.

2.3 Equilibrium statistics on the circle: Bessel functions replace Langevin

$U = -\mathbf{m}\cdot\mathbf{H} = -kT\,\xi\cos\phi$, $\xi = mH/kT$; $W_0 \propto e^{\xi\cos\phi}$.

$$Z = \int_0^{2\pi} e^{\xi\cos\phi}\mathrm{d}\phi = 2\pi I_0(\xi), \qquad \langle\cos\phi\rangle_0 = \frac{Z'}{Z} = \frac{I_1(\xi)}{I_0(\xi)} \equiv L_{\mathrm{2D}}(\xi), \tag{6}$$

with I_n the modified Bessel functions. So

$$\boxed{\mathbf{M}_0 = nm\,L_{\mathrm{2D}}(\xi)\,\mathbf{h}, \qquad L_{\mathrm{2D}} = \frac{I_1}{I_0}} \qquad L_{\mathrm{2D}} \simeq \frac{\xi}{2} - \frac{\xi^3}{16}\ (\xi \ll 1), \quad L_{\mathrm{2D}} \to 1 - \frac{1}{2\xi}\ (\xi \gg 1), \tag{7}$$

hence the 2D initial susceptibility $\chi_2 = nm^2/2kT$ (vs. $nm^2/3kT$: in 2D no moment "wastes" alignment out of plane). Second harmonic, via the Bessel recurrence $I_{n-1}(\xi) - I_{n+1}(\xi) = \frac{2n}{\xi} I_n(\xi)$ with $n = 1$:

$$\langle\cos 2\phi\rangle_0 = \frac{I_2}{I_0} = 1 - \frac{2L_{\mathrm{2D}}}{\xi}, \qquad \langle\cos^2\phi\rangle_0 = 1 - \frac{L_{\mathrm{2D}}}{\xi}, \quad \langle\sin^2\phi\rangle_0 = \frac{L_{\mathrm{2D}}}{\xi}, \tag{8}$$

(recurrence verified numerically to 10^{-16}), and the 2D orientation tensor under a Gibbs state aligned with $\hat{\boldsymbol{\nu}}$:

$$\boxed{\langle\mathbf{ee}\rangle_0 = \frac{L_{\mathrm{2D}}}{\xi}\mathbb{K}_2 + \Big(1 - \frac{2L_{\mathrm{2D}}}{\xi}\Big)\hat{\boldsymbol{\nu}}\hat{\boldsymbol{\nu}}} \qquad (\mathrm{tr} = \tfrac{2L_{\mathrm{2D}}}{\xi} + 1 - \tfrac{2L_{\mathrm{2D}}}{\xi} = 1\ \checkmark). \tag{9}$$

Compare 3D: $\langle\mathbf{ee}\rangle = \frac{L}{\xi}\mathbb{K}_3 + (1 - \frac{3L}{\xi})\hat{\boldsymbol{\nu}}\hat{\boldsymbol{\nu}}$ — the coefficient $3 \to 2$ is the trace/dimension count. A further identity we will need (analog of $L' = 1 - 2L/\xi - L^2$):

$$L'_{\mathrm{2D}}(\xi) = 1 - \frac{L_{\mathrm{2D}}}{\xi} - L_{\mathrm{2D}}^2 \tag{10}$$

proved from $I_1' = \frac{1}{2}(I_0 + I_2) = I_0 - I_1/\xi$ and $I_0' = I_1$; verified numerically to 10^{-10}.

The Néel-vs-Brownian discussion is unchanged from 3D (it concerns the grain interior): $\sigma = KV/kT \gg 1$ freezes the internal channel and makes the rigid-planar-rotor model legitimate, now with the extra requirement of easy-*plane* confinement of $\mathbf{m}$ (Sec. 1).

3 Fokker–Planck on the circle and the *exactly solvable* moment hierarchy

3.1 The kinetic equation

Torque balance on one rotor (inertia dropped): $\omega_{\mathrm{det}} = \Omega + \zeta_r^{-1}(\mathbf{m}\times\mathbf{H})_z = \Omega - \zeta_r^{-1}\partial_\phi U$, since $(\mathbf{m}\times\mathbf{H})_z = mH\sin(0-\phi) = -\partial_\phi U$ for $\mathbf{h} = \hat{\mathbf{x}}$. Probability conservation on S^1 with diffusion $D_r = kT/\zeta_r$:

$$\boxed{\frac{\partial W}{\partial t} = \frac{\partial}{\partial\phi}\Big[D_r\Big(\frac{\partial W}{\partial\phi} + W\,\frac{\partial}{\partial\phi}\frac{U}{kT}\Big) - \Omega\,W\Big]}, \qquad \frac{U}{kT} = -\xi(t)\cos\phi. \tag{11}$$

Checks: Gibbs state annihilates the bracket at $\Omega = 0$; $\int_0^{2\pi}\partial_\phi(\cdot) = 0$ conserves norm; uniform W is stationary under pure Ω. ✓

3.2 Fourier moments: a closed *tridiagonal* hierarchy

Define $m_n(t) = \langle e^{in\phi}\rangle$, $m_0 = 1$, $m_{-n} = m_n^*$. Multiply (11) by $e^{in\phi}$, integrate by parts twice, and use $\int e^{in\phi}\sin\phi\, W\,\mathrm{d}\phi = (m_{n+1} - m_{n-1})/2i$:

$$\boxed{\tau_B \frac{\mathrm{d}m_n}{\mathrm{d}t} = in\Omega\tau_B\, m_n \;-\; n^2 m_n \;+\; \frac{n\,\xi(t)}{2}\big(m_{n-1} - m_{n+1}\big)} \tag{12}$$

(time in units of $\tau_B = 1/D_r$; the $-n^2$ establishes the $n^2 D_r$ relaxation law claimed in Sec. 2). This is the decisive structural difference from 3D: on the sphere the analogous hierarchy couples spherical harmonics with l- and m-dependent coefficients; on the circle it is a *simple tridiagonal chain*, i.e. the 2D Fokker–Planck problem is numerically *exactly* solvable at trivial cost. We exploit this throughout to benchmark every closure.

Equilibrium check of (12)*:* stationarity at $\Omega = 0$, constant ξ, requires $n^2 m_n = \frac{n\xi}{2}(m_{n-1} - m_{n+1})$ with $m_n = I_n/I_0$ — which is *exactly* the Bessel recurrence $I_{n-1} - I_{n+1} = \frac{2n}{\xi} I_n$. Every coefficient of (12) is thereby verified at once. ✓

3.3 Linearization in $\Omega\tau_B$ (the form used for all numerics)

Write $m_n = a_n + i\,\Omega\tau_B\, b_n$ with a_n, b_n real (W is even in ϕ at $\Omega = 0$). Real and imaginary parts of (12):

$$\tau_B \dot{a}_n = -n^2 a_n + \tfrac{n\xi(t)}{2}(a_{n-1} - a_{n+1}), \qquad \tau_B \dot{b}_n = n\,a_n - n^2 b_n + \tfrac{n\xi(t)}{2}(b_{n-1} - b_{n+1}), \tag{13}$$

$a_0 = 1$, $b_0 = 0$. The torque density is $\langle(\mathbf{m}\times\mathbf{H})_z\rangle n = -nmH(t)\langle\sin\phi\rangle = -nmH(t)\,\mathrm{Im}\, m_1$, so with (4) and $nmH\tau_B = n\zeta_r\xi$:

$$\overline{(\mathbf{M}\times\mathbf{H})_z} = -n\zeta_r\,\Omega\,\overline{\xi(t)\,b_1(t)} \;\Rightarrow\; \boxed{g_{\text{exact}} = \overline{\xi(t)\,b_1(t)}}. \tag{14}$$

3.4 Exact first-moment equation and its vector form

The $n = 1$ member of (12) is the exact (unclosed) magnetization equation. In vector notation, using $\frac{1}{2}(1 - m_2) \leftrightarrow \mathbf{h} - \langle\mathbf{ee}\rangle\cdot\mathbf{h}$ (checked component-wise: real part $= \langle\sin^2\phi\rangle$, imaginary part $= -\langle\sin\phi\cos\phi\rangle$):

$$\boxed{\frac{\mathrm{d}\langle\mathbf{e}\rangle}{\mathrm{d}t} = \boldsymbol{\Omega}\times\langle\mathbf{e}\rangle - \frac{1}{\tau_B}\langle\mathbf{e}\rangle + \frac{\xi}{\tau_B}\big(\mathbf{h} - \langle\mathbf{ee}\rangle\cdot\mathbf{h}\big)} \tag{15}$$

Instructive comparison in units of D_r: 3D reads $-2D_r\langle\mathbf{e}\rangle + D_r\xi(\mathbf{h} - \langle\mathbf{ee}\rangle\mathbf{h})$; 2D reads $-D_r\langle\mathbf{e}\rangle + D_r\xi(\mathbf{h} - \langle\mathbf{ee}\rangle\mathbf{h})$. The *potential* coupling is dimension-independent; only the free-relaxation eigenvalue changes ($l(l+1) = 2 \to n^2 = 1$). Equilibrium check with (9): $\langle\mathbf{ee}\rangle\mathbf{h} = (1 - \frac{L_{2D}}{\xi})\mathbf{h}$ gives RHS $= -\frac{L_{2D}}{\tau_B}\mathbf{h} + \frac{\xi}{\tau_B}\frac{L_{2D}}{\xi}\mathbf{h} = 0$. ✓

4 Closures in 2D

4.1 Debye closure (exact to $O(\xi)$)

Near-isotropy: $m_2 \to 0$ ($\langle\mathbf{ee}\rangle \to \mathbb{1}/2$), and (15) becomes, with $\mathbf{M} = nm\langle\mathbf{e}\rangle$, $\chi_2 = nm^2/2kT$:

$$\boxed{\frac{\mathrm{d}\mathbf{M}}{\mathrm{d}t} = \boldsymbol{\Omega}\times\mathbf{M} - \frac{1}{\tau_B}\big(\mathbf{M} - \chi_2\mathbf{H}(t)\big)} \qquad (\xi \ll 1,\ \text{any } \omega), \tag{16}$$

again a theorem, not a postulate. The finite-ξ phenomenological (Shliomis-72-type) extension is $\mathrm{d}\mathbf{M}/\mathrm{d}t = \boldsymbol{\omega}_p\times\mathbf{M} - (\mathbf{M} - nmL_{2D}(\xi(t))\mathbf{h})/\tau_B$, exact for statics, uncontrolled for finite ξ *and* $\omega\tau_B$ — and in 2D we can *measure* how uncontrolled (Sec. 10).

4.2 Effective-field (MRSh-type) closure: the 2D analog of PRL Eq. (6), proved exactly

Ansatz: $W_e \propto e^{\boldsymbol{\xi}_e\cdot\mathbf{e}}$ with planar effective field $\boldsymbol{\xi}_e = \xi_e(\cos\psi, \sin\psi)$, so

$$\boxed{\mathbf{M} = nm\, L_{2\mathrm{D}}(\xi_e)\,\frac{\boldsymbol{\xi}_e}{\xi_e}}, \qquad \langle\mathbf{ee}\rangle_e \text{ from (9) with } \hat{\boldsymbol{\nu}} = \hat{\boldsymbol{\xi}}_e. \tag{17}$$

Substituting into (15) and regrouping (the algebra is done *exactly*, at arbitrary angle ψ, in complex notation $\zeta = \xi_e e^{i\psi}$: both the real and imaginary parts of the two forms agree identically — the check is spelled out in the Ledger) gives the 2D effective-field equation

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\Big(L_e\frac{\boldsymbol{\xi}_e}{\xi_e}\Big) = \boldsymbol{\Omega}\times\Big(L_e\frac{\boldsymbol{\xi}_e}{\xi_e}\Big) - \frac{1}{\tau_B}\frac{L_e}{\xi_e}(\boldsymbol{\xi}_e - \boldsymbol{\xi}) - \frac{1}{\tau_B\,\xi_e^2}\Big(1 - \frac{2L_e}{\xi_e}\Big)\,\boldsymbol{\xi}_e\times(\boldsymbol{\xi}_e\times\boldsymbol{\xi})}} \tag{18}$$

with $L_e = L_{2\mathrm{D}}(\xi_e)$, $\boldsymbol{\xi} = \xi(t)\mathbf{h}$. Dictionary vs. 3D PRL (6): $L \to L_{2\mathrm{D}} = I_1/I_0$; the transverse coefficient $\frac{1}{2}\big(1 - \frac{3L_e}{\xi_e}\big) \to 1\cdot\big(1 - \frac{2L_e}{\xi_e}\big)$; $\tau_B = 1/D_r$. Linearized relaxation times:

$$\tau_\parallel = \tau_B\frac{\mathrm{d}\ln L_{2\mathrm{D}}}{\mathrm{d}\ln\xi}, \qquad \tau_\perp = \tau_B\,\frac{L_{2\mathrm{D}}}{\xi - L_{2\mathrm{D}}} \quad \big(\text{both} \to \tau_B \text{ at } \xi\to 0;\ \to \tau_B/\xi \text{ at } \xi\gg 1\big). \tag{19}$$

Part II
2D ferrohydrodynamics and the viscosity theorem

5 Coupled equations and the theorem $\Delta\eta = \eta\varphi\, g$

5.1 Torque balance and momentum equation

Summing single-rotor torque balances over n particles per area (inertia dropped, $\tau_s/\tau_B \sim 10^{-5}$):

$$\boxed{n\zeta_r(\omega_p - \Omega) = (\mathbf{M}\times\mathbf{H})_z, \qquad n\zeta_r = 4\eta\varphi} \tag{20}$$

— the 2D analog of PRL (1) with $6\eta\varphi \to 4\eta\varphi$. The torque density $T\hat{\mathbf{z}}$, $T = (\mathbf{M}\times\mathbf{H})_z$, transmitted to the fluid appears as the antisymmetric stress $\sigma^a_{ik} = -\frac{1}{2}\varepsilon_{ik}T$ (ε_{ik} the 2D Levi–Civita symbol), whose divergence is $f_i = \frac{1}{2}(\nabla\times T\hat{\mathbf{z}})_i = \frac{1}{2}(\partial_y T, -\partial_x T)$; total angular momentum (orbital + internal) is conserved by the same bookkeeping as in 3D. Hence

$$\rho\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = -\nabla p + (\mathbf{M}\cdot\nabla)\mathbf{H} + \eta\Delta\mathbf{v} + \tfrac{1}{2}\nabla\times(T\hat{\mathbf{z}}). \tag{21}$$

5.2 The theorem

For 2D incompressible flow, $\Delta\mathbf{v} = -\nabla\times(\nabla\times\mathbf{v}) = -2\nabla\times(\Omega\hat{\mathbf{z}})$. If the field-period-averaged torque obeys $\overline{T} = -4\eta\varphi\, g\,\Omega$ [cf. (20)], then

$$\eta\Delta\mathbf{v} + \tfrac{1}{2}\nabla\times(\overline{T}\hat{\mathbf{z}}) = -2\eta(1+\varphi g)\,\nabla\times(\Omega\hat{\mathbf{z}}), \tag{22}$$

i.e. ordinary 2D Navier–Stokes with viscosity $\eta + \Delta\eta$:

$$\boxed{\Delta\eta = \eta\varphi\, g = \eta\varphi\,\frac{\Omega - \omega_p}{\Omega}}, \qquad \omega_p = (1-g)\Omega, \tag{23}$$

using $\omega_p = \Omega + \overline{T}/n\zeta_r = (1-g)\Omega$ from (20). Compare 3D: $\Delta\eta = \frac{3}{2}\eta\varphi\, g$. The DC thermodynamic bound holds verbatim (a magnet does no work): $0 \le g \le 1$, so $0 \le \Delta\eta \le \eta\varphi$ for static fields; saturation $\Delta\eta_{\max} = \eta\varphi$ (vs. $\frac{3}{2}\eta\varphi$).

Part III
The three field regimes in 2D

6 Static field: two closures *and the exact answer*

6.1 Debye/Shliomis-type closure

Steady state, $\mathbf{h} \perp$ effective rotation as in 3D, $M_y = \omega_p \tau_B M_x$, $M_x \simeq M_0 = nmL_{2D}(\xi)$; torque $-\omega_p \tau_B M_0 H$; with $\tau_B M_0 H = n\zeta_r\, \xi L_{2D}$ inserted into (20):

$$\omega_p = \frac{\Omega}{1+\xi L_{2D}} \;\Rightarrow\; \boxed{\Delta\eta^{\mathrm{Deb}} = \eta\varphi\, \frac{\xi L_{2D}(\xi)}{1+\xi L_{2D}(\xi)}} \qquad \Big(\to \tfrac{1}{2}\eta\varphi\xi^2,\ \xi \ll 1;\ \to \eta\varphi,\ \xi \gg 1\Big). \tag{24}$$

6.2 Effective-field closure

Transverse balance of (18) at constant ξ, first order in $\Omega\tau_B$ (algebra identical in structure to 3D): restoring rate $= \frac{1}{\tau_B}\big[\frac{L_{2D}}{\xi} + \big(1 - \frac{2L_{2D}}{\xi}\big)\big] = \frac{\xi - L_{2D}}{\tau_B \xi}$, drive $L_{2D}\Omega$, hence $g = \xi L_{2D} F$ with $F = L_{2D}/(\xi - L_{2D})$:

$$\boxed{\Delta\eta^{\mathrm{EF}} = \eta\varphi\, \frac{\xi L_{2D}^2(\xi)}{\xi - L_{2D}(\xi)}} \qquad \text{(same limits)}. \tag{25}$$

6.3 The exact static g (unavailable in closed form in 3D, trivial here)

Stationary linear response of the hierarchy (13): $a_n = I_n(\xi)/I_0(\xi)$ exactly, and b_n solves the *tridiagonal linear system*

$$n^2 b_n - \frac{n\xi}{2}\big(b_{n-1} - b_{n+1}\big) = n\, a_n, \qquad b_0 = 0, \qquad g_{\mathrm{exact}} = \xi\, b_1. \tag{26}$$

Solved numerically (truncation $N = 80$; results N-independent to 10^{-6}):

ξ	0.5	1	2	3	5	8	12
g_{exact}	0.1158	0.3761	0.8076	0.9580	0.9987	1.0000	1.0000
$g^{\mathrm{EF}} = \xi L_{2D}^2/(\xi - L_{2D})$	0.1142	0.3599	0.7478	0.8987	0.9718	0.9905	0.9960
$g^{\mathrm{Deb}} = \xi L_{2D}/(1+\xi L_{2D})$	0.1081	0.3086	0.5826	0.7085	0.8171	0.8821	0.9199

Findings: (i) the exact g respects the thermodynamic bound $g \le 1$ and saturates *much faster* than either closure; (ii) the effective-field closure is clearly the better one (6% low at $\xi = 3$) but the Debye closure is off by up to 26%; (iii) in 2D one should simply use (26) — it is exact and costs a tridiagonal solve.

7 Weak alternating field: universal shape, 2D prefactors

$\xi_0 \ll 1$: Debye equation (16) is exact, so this regime is closure-independent. The algebra is *identical* in form to 3D (a two-pole linear filter):

$$M_x = \chi_2 H_0 \cos\alpha\, \cos(\omega t - \alpha), \qquad \tan\alpha = \omega\tau_B, \tag{27}$$

$$M_y = \Omega\tau_B\, \chi_2 H_0 \cos^2\alpha\, \cos(\omega t - 2\alpha), \tag{28}$$

mean torque $\overline{T} = -\frac{1}{2}\Omega\tau_B\chi_2 H_0^2\cos^2\alpha\cos 2\alpha$, and now $\tau_B\chi_2 H_0^2 = \frac{\zeta_r}{kT}\frac{nm^2}{2kT}H_0^2 = \frac{n\zeta_r}{2}\xi_0^2$, so $g = \frac{\xi_0^2}{4}\cos^2\alpha\cos 2\alpha$ and

$$\boxed{\Delta\eta(\omega) = \frac{1}{4}\eta\varphi\,\xi_0^2\,\frac{1-\omega^2\tau_B^2}{(1+\omega^2\tau_B^2)^2} = \frac{1}{2}\eta\varphi\,\bar{\xi}^{\,2}\,\frac{1-\omega^2\tau_B^2}{(1+\omega^2\tau_B^2)^2}}, \qquad \bar{\xi} = \frac{\xi_0}{\sqrt{2}}. \tag{29}$$

Twice the 3D amplitude-convention result $\frac{1}{8}\eta\varphi\xi_0^2 R$. Factor bookkeeping (each piece verified): $g_{2D}/g_{3D} = 3$, composed of $\chi_2/\chi_3 = \frac{3}{2}$ and $\tau_B^{2D}/\tau_B^{3D} = 2$ at fixed ζ_r; the theorem prefactors ($\eta\varphi$ vs $\frac{3}{2}\eta\varphi$) contribute $\frac{2}{3}$; net $3\times\frac{2}{3} = 2$. *Universal statements, dimension-independent:*

- shape function $R(x) = (1-x^2)/(1+x^2)^2$ identical; sign change at $\omega\tau_B = 1$; minimum at $\omega\tau_B = \sqrt{3}$ of depth $-\Delta\eta(0)/8$; high-frequency tail $\propto -\omega^{-2}$;
- but τ_B itself differs: for the same sphere with easy-plane moments, $\tau_B^{2D} = 2\tau_B^{3D}$, so the crossover frequency $f_0 = 1/2\pi\tau_B$ is *halved* — an experimentally testable 2D fingerprint.

Mean particle spin: $\Theta = \big[1 - \frac{\xi_0^2}{4}R(\omega\tau_B)\big]\Omega$; inserting into (23) reproduces (29) identically. ✓
Numerical validation: the exact hierarchy (13) at $\xi_0 = 0.3$ returns $g = \frac{\xi_0^2}{4}R$ within 2% at $\omega\tau_B = 0.3$ and to 4 digits at $\omega\tau_B = \sqrt{3}, 3$, with the exact zero at $\omega\tau_B = 1$; the effective-field ODEs give the same. ✓

8 Arbitrary amplitude, Debye closure; the square-wave solution

With $\xi(t) = \xi\gamma(t)$, $M_x = nmF$, $M_y = nm\Omega\tau_B G$, projecting the phenomenological equation (nonlinear torque term worked out exactly as in 3D, but the 2D coefficient is 1, not $\frac{1}{2}$):

$$\tau_B\dot{F} = L_{2D}[\xi\gamma(t)] - F, \qquad \tau_B\dot{G} = F - \big(1+\xi\gamma F\big)G, \qquad \boxed{g = \xi\,\overline{G\gamma}, \quad \Delta\eta = \eta\varphi\,\xi\,\overline{G\gamma}} \tag{30}$$

and averaging the torque balance gives, at all amplitudes, $\xi\overline{G\gamma} = (\Omega - \overline{\Theta})/\Omega$ — the 2D version of the SM structural theorem. For the *weak-field square wave* the linearized system ($L_{2D} \to \frac{\xi}{2}\gamma$; drop the nonlinear factor) is the same two-stage filter as in 3D with $a = \xi/2$; the derivation of the earlier document carries over verbatim and yields

$$\boxed{\Delta\eta = \frac{1}{2}\eta\varphi\,\xi^2\Big(2 - \tanh^2\varepsilon - \frac{2\tanh\varepsilon}{\varepsilon}\Big), \qquad \varepsilon = \frac{\pi}{2\omega\tau_B}} \tag{31}$$

— the *same* shape function as in 3D (hence the same $\omega_0\tau_B = 0.978$, $\omega_m\tau_B = 1.708$, depth $-0.105\,\Delta\eta(0)$), with the 2D prefactor $\frac{1}{2}$ (static check: $\varepsilon\to\infty$ gives $\frac{1}{2}\eta\varphi\xi^2$, the 2D weak-field static value for $\gamma^2 \equiv 1$ ✓). The frequency-shape universality is a genuine theorem: at $O(\xi^2)$ the response is a dimension-independent linear filter; dimensionality enters only through χ, $n\zeta_r$, the theorem prefactor, and the definition of τ_B.

Part IV
The 2D "Bacri theory" and its exact benchmark

9 Perturbation theory in $\Omega\tau_B$ for the 2D effective-field equation

Expand $\boldsymbol{\xi}_e = \xi_e(t)\mathbf{h} + \boldsymbol{\delta}$, $\boldsymbol{\delta}\perp\mathbf{h} = O(\Omega\tau_B)$, $\boldsymbol{\xi} = \xi_0\cos\omega t\,\mathbf{h}$.

Zeroth order [2D analog of PRL (7)] — project (18) on $\mathbf{h}$:

$$\boxed{\tau_B\frac{\mathrm{d}\xi_e}{\mathrm{d}t} = -\Big(\frac{\mathrm{d}\ln L_{2D}}{\mathrm{d}\xi_e}\Big)^{-1}\Big(1 - \frac{\xi_0}{\xi_e}\cos\omega t\Big)}, \qquad \Big(\frac{\mathrm{d}\ln L_{2D}}{\mathrm{d}\xi_e}\Big)^{-1} = \frac{L_{2D}}{L'_{2D}}\ \big[\text{use (10)}\big], \tag{32}$$

identical in form to 3D; ξ_e is a signed variable, regular through zero ($L_{2D}/L'_{2D} \to \xi_e$).

First order [2D analog of PRL (8)] — with the ansatz $\mathbf{M}^{(1)} = nm\, L_{2D}(\xi_e)\, F\, \tau_B\, \boldsymbol{\Omega} \times \mathbf{h}$, the same sequence of steps as in 3D (transverse projection; substitute the zeroth-order identity $\tau_B \dot{L}_e = -L_e(1 - \xi_0 \cos\omega t/\xi_e)$; divide by L_e) gives

$$\boxed{\tau_B \frac{\mathrm{d}F}{\mathrm{d}t} = 1 - \left(\frac{1}{L_{2D}(\xi_e)} - \frac{1}{\xi_e}\right) \xi_0 F \cos\omega t} \tag{33}$$

— note: *no* factor $\frac{1}{2}$ (it traces back to $1 - \frac{2L_e}{\xi_e}$ vs $\frac{1}{2}(1 - \frac{3L_e}{\xi_e})$).

Torque [2D analog of PRL (9)]: $\mathbf{M}^{(1)} \times \mathbf{H} = -nmH_0 \cos\omega t\, L_{2D} F \tau_B\, \boldsymbol{\Omega} = -n\zeta_r\, \xi_0 \cos\omega t\, L_{2D} F\, \boldsymbol{\Omega}$, hence

$$\boxed{g = \xi_0\, \overline{\cos\omega t\; L_{2D}(\xi_e)\, F}\;, \qquad \Delta\eta = \eta\varphi\, g, \qquad \omega_p = (1-g)\Omega} \tag{34}$$

(again no $\frac{1}{2}$). *Regular numerical form:* $P \equiv L_{2D}(\xi_e)F$ obeys $\tau_B \dot{P} = L_{2D}(\xi_e) - P - P\, s_2(\xi_e)\, \xi_0 \cos\omega t$ with the smooth odd function $s_2(\xi) \equiv \frac{1}{L_{2D}(\xi)} - \frac{2}{\xi} = \frac{\xi}{4} + O(\xi^3)$, and $g = \xi_0 \overline{\cos\omega t\, P}$.

Static check: $\dot{F} = 0 \Rightarrow F = L_{2D}/(\xi - L_{2D})$, $g = \xi L_{2D}^2/(\xi - L_{2D})$ — the EF static formula of Sec. 6. ✓ *Weak-field check:* numerics reproduce $g = \frac{\xi_0^2}{4} R(\omega\tau_B)$. ✓ *Independent cross-check of the reduction:* direct integration of the *full* complex equation (18) at small $\Omega\tau_B = 0.01$ gives $g(5,2) = -0.427$, identical to the perturbative ODEs. ✓

10 Benchmark: effective-field closure vs. exact Fokker–Planck (the 2D payoff)

Because (13) is exact and cheap, we can do in 2D what nobody has done for the 3D driven problem: measure the closure error. RK4 integration of both systems to the periodic attractor (convergence verified in truncation N, step size, and cycle count; e.g. $g_{\mathrm{exact}}(10, 1.5) = 0.1578$ stable across $N = 35, 45, 55$):

g: exact FP / effective field	$\xi_0 = 2$	$\xi_0 = 5$	$\xi_0 = 10$
$\omega\tau_B = 0.5$	0.295 / 0.286	0.540 / 0.512	0.628 / 0.584
$\omega\tau_B = 1.0$	0.038 / 0.023	0.238 / 0.160	0.379 / 0.248
$\omega\tau_B = 1.5$	−0.077 / −0.098	−0.026 / −0.177	**+0.158** / **−0.070**
$\omega\tau_B = 2.0$	−0.100 / −0.113	−0.220 / −0.427	−0.048 / −0.373

Conclusions (all three codes cross-validated in their common limits, so these are properties of the *closure*, not of the numerics):

1. At weak drive or low frequency the closure is decent ($\lesssim 10\%$).
2. At strong drive near and above the crossover ($\xi_0 \gtrsim 5$, $\omega\tau_B \gtrsim 1$) it fails *quantitatively by factors of 2* and, at $(\xi_0, \omega\tau_B) = (10, 1.5)$, even *predicts the wrong sign* of $\Delta\eta$.
3. Therefore in 2D the correct procedure for the "Bacri problem" is: solve the exact hierarchy (13) (tridiagonal, trivial). The effective-field machinery, essential in 3D, is dispensable — and demonstrably unreliable — in 2D.
4. **Caution exported back to 3D:** the one-mode Gibbs closure has never been benchmarked against the full FP for the strongly driven AC problem in 3D either; the 2D result suggests its 3D errors in the deep negative-viscosity regime may be larger than the few-percent lore (which was established for relaxation, not for strong periodic driving). This is a concrete research question.

Qualitatively, however, the exact 2D solution reproduces the full Bacri phenomenology: $g > 0$ for $\omega\tau_B < 1$; sign change near $\omega\tau_B \simeq 1$; nonmonotonic negative minima at higher frequency; recovery toward $g \to 0^-$ (with structure) at large amplitude.

11 Zero total viscosity in 2D

$\eta_{\text{total}} = 0$ requires $\Delta\eta = -\eta$, i.e. by (23)

$$g = -\frac{1}{\varphi} \qquad (\text{vs. } -\tfrac{2}{3\varphi} \text{ in 3D}), \tag{35}$$

e.g. $g = -5$ at $\varphi = 0.2$. Exact-FP scan at large drive: $g(10, 3) = -0.409$, $g(20, 3) = -0.176$, $g(20, 5) = -0.729$ — the data are consistent with $|g|$ saturating at the DC scale ~ 1 along the large-$(\xi_0, \omega\tau_B)$ diagonal, exactly as in the 3D isoline maps. Hence $\eta_{\text{total}} = 0$ would need $\varphi \gtrsim 1/|g|_{\max} \approx 1$, far outside the dilute single-rotor framework: *unattainable in 2D for the same reason as in 3D, but with an even cleaner numerical demonstration.* What survives is the practical result: a frequency knob tunes the 2D suspension viscosity continuously between $\eta(1 + \varphi g^{+}_{\max})$ and $\eta(1 + \varphi g^{-}_{\min})$, i.e. roughly $\eta(1 \pm \varphi \cdot O(1))$.

12 2D vs. 3D dictionary

Quantity	3D	2D
Orientation space	sphere S^2	circle S^1
Friction coefficient	$\zeta_r = 8\pi\eta a^3 = 6\eta V$	$\zeta_r = 4\pi\eta a^2$ (strict 2D)
Friction density	$n\zeta_r = 6\eta\varphi$	$n\zeta_r = 4\eta\varphi$
Brownian time	$\tau_B = 1/2D_r = 3\eta V/kT$	$\tau_B = 1/D_r = \zeta_r/kT$
Equilibrium law	$L(\xi) = \coth\xi - 1/\xi$	$L_{2\text{D}}(\xi) = I_1(\xi)/I_0(\xi)$
Initial susceptibility	$nm^2/3kT$	$nm^2/2kT$
$\langle \mathbf{ee} \rangle$ anisotropy coeff.	$1 - 3L/\xi$	$1 - 2L_{2\text{D}}/\xi$
Derivative identity	$L' = 1 - 2L/\xi - L^2$	$L'_{2\text{D}} = 1 - L_{2\text{D}}/\xi - L^2_{2\text{D}}$
Free relaxation eigenvalue	$l(l+1)D_r$	$n^2 D_r$
FP moment hierarchy	spherical-harmonic couplings	*tridiagonal*, exactly solvable
EF transverse coefficient	$\frac{1}{2}(1 - 3L_e/\xi_e)$	$1 - 2L_e/\xi_e$
F-equation coefficient	$\frac{1}{2}(1/L_e - 1/\xi_e)$	$1/\underline{L_e} - 1/\xi_e$
g quadrature	$\frac{1}{2}\xi_0 \overline{\cos\omega t\, LF}$	$\xi_0 \overline{\cos\omega t\, L_{2\text{D}} F}$
Viscosity theorem	$\Delta\eta = \frac{3}{2}\eta\varphi g$	$\Delta\eta = \eta\varphi g$
DC saturation	$\frac{3}{2}\eta\varphi$	$\eta\varphi$
Weak-AC amplitude prefactor	$\frac{1}{8}\eta\varphi\xi_0^2$	$\frac{1}{4}\eta\varphi\xi_0^2$
Shapes $R(\omega\tau_B)$, $f(\varepsilon)$	identical:	universal at $O(\xi^2)$
Sign-change criterion	$\omega\tau_B = 1$	$\omega\tau_B = 1$; τ_B doubled for same particle
Zero-η_{total} condition	$g = -2/3\varphi$	$g = -1/\varphi$

13 Verification ledger

Result	Verification performed
$\zeta_r = 4\pi\eta a^2$; $n\zeta_r = 4\eta\varphi$	explicit 2D Stokes rotlet solution; stress integrated; no-paradox argument
$\tau_B = 1/D_r$	$n^2 D_r$ law read off the exact hierarchy (12)
$L_{2D} = I_1/I_0$; $\langle\cos 2\phi\rangle = 1 - 2L_{2D}/\xi$; (9); (10)	Bessel recurrence and derivative identity checked to $10^{-16}/10^{-10}$; trace and both limits of (9)
FP equation (11)	Gibbs annihilation, norm, isotropy checks
Hierarchy (12)	equilibrium stationarity = Bessel recurrence, verifying every coefficient at once
Exact moment eq. (15)	complex↔vector map checked component-wise; equilibrium reproduced
Debye closure (16)	exact $O(\xi)$ limit of (15)
2D EF equation (18)	proved *exactly* (all ψ): real and imaginary parts of both forms matched analytically
$\tau_{\parallel}, \tau_{\perp}$	both $\to \tau_B$ at $\xi \to 0$
Theorem $\Delta\eta = \eta\varphi g$	stress grouping; independently via averaged torque balance; DC bound $g \le 1$ confirmed by exact statics
Static g: exact/EF/Debye	tridiagonal solve, N-convergence to 10^{-6}; limits; table
Weak-AC (29)	quasi-static factor check; exact-FP and EF numerics at $\xi_0 = 0.3$ ($\le 2\%$, exact zero at $\omega\tau_B = 1$)
Square-wave result	derivation carried over; static limit check; shape universality argued at $O(\xi^2)$
2D (7′),(8′),(9′)	static and weak-field limits recovered; perturbative ODEs = full complex equation at $(5, 2)$ to 3 digits
Closure benchmark table	three independent codes cross-validated in shared limits; N/step/cycle convergence at the discrepant point
Negative-g scan	exact FP; consistency with $\lvert g\rvert \lesssim 1$ saturation

14 Main structural conclusions and open problems

1. Every step of the 3D chain (FP → exact moment equation → closures → torque balance → viscosity theorem → three field regimes) has an exact 2D counterpart; the boxed dictionary gives the complete translation. The negative-viscosity effect, its mechanism, and the $\omega\tau_B = 1$ criterion survive intact in 2D.
2. The weak-field frequency dependence (R and the square-wave f) is *dimension-independent*; dimensionality enters only via prefactors and the value of τ_B. Prediction: confining moments to a plane halves the sign-change frequency of the same colloid.
3. 2D bonus: the Fokker–Planck problem is exactly solvable (tridiagonal), which (i) supplies the exact static $\Delta\eta(\xi)$ in one line of linear algebra, and (ii) exposes a serious failure of the effective-field closure under strong AC drive — including sign errors — recommending the exact hierarchy as the standard tool in 2D and casting a documented doubt on untested closure accuracy in the corresponding 3D regime.
4. Open: (a) 3D FP benchmark of the MRSh closure in the driven regime (spherical-harmonic hierarchy; the direct 3D analog of Sec. 10); (b) dipolar interactions/chaining in 2D at realistic φ; (c) rotating in-plane fields in 2D (angular-momentum transfer at all frequencies); (d) membrane experiments: magnetic disks in lipid bilayers as a clean quasi-2D test of the halved crossover frequency.